

Problem Set-2

1. $a^{[1]} \rightarrow 128$ vector.

$$a_B^{[1]} = \begin{bmatrix} a^{[1]} \\ 1 \end{bmatrix}$$

$$\nabla_{a^{[1]}} (a_B^{[1]}) = \nabla_{a^{[1]}} \begin{bmatrix} a^{[1]} \\ 1 \end{bmatrix} = \begin{bmatrix} \nabla_{a^{[1]}} a^{[1]} \\ \nabla_{a^{[1]}} 1 \end{bmatrix}$$

$$z = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

2. $z = 128$ vector

$$a = \text{ReLU}(z).$$

$$\nabla_z(a) = \nabla_z(\text{ReLU}(z)) =$$

3. $a = 784$ vector

One hot encoded word label vector $y \Rightarrow 10$ vector

$$x_B = \begin{bmatrix} x \\ 1 \end{bmatrix} = z^{[1]} = W_B^{[1]} x_B \rightarrow a^{[1]} = \text{ReLU}(z^{[1]}).$$

(128) (128) (128 \times 785) (785)

$$a_B^{[1]} = \begin{bmatrix} a^{[1]} \\ 1 \end{bmatrix} \Rightarrow z^{[2]} = W^{[2]} a_B^{[1]}$$

(10) (10 \times 129) (129)

$$\Rightarrow a^{[2]} = \text{softmax}(z^{[2]})$$

(10) (10)

$$4. \nabla_{a^{(1)}}(L) = \nabla_{a^{(1)}}(a_B^{(1)}) \times \nabla_{a_B^{(1)}}(z^{(2)}) \times \nabla_{z^{(2)}}(L)$$

$$\nabla_{z^{(2)}}(L) = \nabla_{z^{(1)}}(a^{(1)}) + \nabla_{a^{(1)}}(L)$$

$$5. \nabla_{a_B^{(1)}}(z^{(2)}) = 0$$

$$\nabla_{z^{(2)}}(a^{(1)}) = 0.$$